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The Kervaire conjecture via banach fixed points in power series groups

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Abstract
We use the Banach Fixed Point Theorem to solve equations in a group within a Magnus power series ring.

Keywords Power series · Magnus ring · Group ring

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1 Introduction

The Kervaire conjecture states that for a group G, adding a single generator and a single relator cannot result in the trivial group. That is $\bar{G} = G * \langle t \rangle / \langle\langle W \rangle\rangle \neq 1$. The stronger form of this conjecture asserts that $G \to \bar{G}$ is injective when the exponent sum $\#_t(W)$ of t in W is nonzero. Note that $\bar{G}$ is obviously nontrivial when $\#_t(W) = 0$ since then $\bar{G}/\langle\langle \mathrm{im}(G) \rangle\rangle \cong \mathbb{Z}$.

There are several beautiful partial results towards the Kervaire conjecture, always proving this strongest form. It is true when G is residually finite [2], when G is locally indicable [3], and when G is torsion-free [4]. A more recent result gives the conjecture when G is hyperlinear [5].

One viewpoint on the Kervaire conjecture, is to view G as lying within a larger group G', and find a solution to $W(t) = 1$ for some $t \in G'$. We then say that the equation $W(t) = 1$ is *solved in the overgroup* G'.

For instance, when G is finite, one views G as a subgroup of $G' = U(n)$ for $n = |G|$. Then a solution t is found by using that the map $G' \to G'$ given by $t \mapsto W(t)$ has degree $\#_t(W) \neq 0$, and must therefore be surjective, and hence $W(t) = 1$ for some t. Thus $W(t)$ is solved in the overgroup $U(n)$.

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In this note, we show that an equation over a right-angled Artin group G can be solved within the associated Magnus power series G'. More generally, this holds for any subgroup of the group of units of such a power series ring.

Subgroups of these groups of units are well-known to be residually finite, torsion-free and orderable. So this case of the Kervaire conjecture was already a consequence of each of the theorems referenced earlier. However, the solution within the power series overgroup is new. More significantly, the mode of proof utilizing the Banach fixed point theorem to solve group-theoretic equations is of interest here, as it offers a new paradigm and hopefully there will be further generalizations along this direction. Other approaches to the Kervaire conjecture using fixed points are given in [1].

2 Power series over monoid

2.1 $\mathbb{R}$-graded monoid

The monoid M is $\mathbb{R}$-*graded* if there is a map $\deg : M \to \mathbb{R}$ such that:

1. $\deg(m) \geq 0$ with equality iff $m = 1$
2. $\deg(mn) = \deg(m) + \deg(n)$

Let R be a ring. Let $R[[M]]$ be the ring consisting of formal sums $\sum_m r_m m$ with $r_m \in R$ for each m and having the property that for each $n \in \mathbb{R}$ we have: $r_m = 0$ for all but finitely m with $\deg(m) \leq n$. Multiplication and addition are defined as usual, and the degree map extends via $\deg$ to $R[[M]]$ by $\deg(r) = \inf\big(\deg(m) : r_m \neq 0\big)$, so $\deg(0) = \infty$.

The classical case introduced by Magnus has $M = M(X)$ the free monoid on $X = \{x_1, \ldots, x_s\}$, and $R = \mathbb{Z}$. This was generalized to another case of great interest, also using $R = \mathbb{Z}$, where $M = M(\Gamma)$ is the partially commutative monoid whose generators correspond to the vertices of the graph Γ, where two generators commute iff they correspond to adjacent vertices. In these two cases $\deg(m)$ is the usual degree of m regarded as a monomial in the variables (i.e. the generators).

2.2 Metric

Define a metric on $R[[M]]$ by:

$$\mathsf{d}(x, y) = 2^{-\deg(x-y)}$$

Observe that $(R[[M]], \mathsf{d})$ is complete, and $R \subset R[[M]]$ is discrete.

We shall later use that $\mathsf{d}(mx, my) = 2^{-\deg(m)}\mathsf{d}(x, y)$ for $m \in M$.

For use in Sect. 4 where $R[[M]]^{\oplus n}$ is a free module of rank $n \in \mathbb{N}$, we extend the notion of degree by

$$\deg(m_1, ..., m_n) := \min_{i=1}^{n} \deg(m_i)$$

and similarly define a metric on $R[[M]]^{\oplus n}$ by $\mathsf{d}(\vec{x}, \vec{y}) = 2^{-\deg(\vec{x}-\vec{y})}$. This is unambiguous as the first metric we defined on $R[[M]]$ agrees with the metric induced by considering $R[[M]]$ as a free module of rank 1 over itself.

3 Main

Definition 1 Let K be the ideal in $R[[M]]$ consisting of elements with zero constant term, i.e. $K = \{r \in R[[M]] : r_1 = 0\}$ where $1 = 1_M$.

Let $G \hookrightarrow (1 + K) \subset R[[M]]^*$ be some arbitrary group embedding into the units of $R[[M]]$ in a neighborhood of 1 and consider a word W in $G * \langle t \rangle$:

$$W = g_1 t^{\epsilon_1} \cdots g_p t^{\epsilon_p}$$

Define $T : K \to K$ by

$$T(k) = (1 + k) - W(1 + k).$$

Remark 2 If $T(s) = s$ then $W(1 + s) = 1$. Indeed,

$$1 \;=\; (1 + s) - s \;=\; (1 + s) - T(s) \;=\; W(1 + s).$$

Theorem 3 (Banach Fixed Point Theorem) *Let (K, d) be a complete metric space. Let $T : K \to K$ satisfy $\mathsf{d}\big(T(a), T(b)\big) \leq \mu \mathsf{d}(a, b)$ for some $\mu < 1$. Then there is a unique $s \in K$ with $T(s) = s$.*

Notation 4 *($*\alpha * \beta*$) Within the sums below we shall use the notation $*\alpha * \beta*$ to indicate a product of terms, where at least one of the terms equals α and at least one of the terms equals β (in either order).*

Theorem 5 *Let $G \subset 1 + K$ be a subgroup of $R[[M]]^*$. Let $W = g_1 t^{\epsilon_1} g_2 t^{\epsilon_2} \cdots g_p t^{\epsilon_p}$ be an equation over G with $\#_t(W) = 1$. Then $W(t) = 1$ has a unique solution in $1 + K$.*

Proof Let $g_i = (1 + \gamma_i)$ for each i. Let $\sigma = \min(\mathsf{deg}(\gamma_i))$.

If $\sigma = \infty$ then each $\gamma_i = 0$, so $W = t$ and hence $0 \in K$ is the unique solution to $T(k) = k$. Indeed, $T(k) = (1 + k) - W(1 + k) = (1 + k) - (1 + k) = 0$.

We now assume some $\gamma_i \neq 0$ and $0 < \sigma < \infty$.

For $k \in K$, observe that

$$\begin{aligned} W(1 + k) \;&=\; (1 + \gamma_1)(1 + k)^{\epsilon_1} \cdots (1 + \gamma_p)(1 + k)^{\epsilon_p} \\ &= -1 \;+ (1 + k)^1 \;+ \prod_{i=1}^{p} (1 + \gamma_i) + \;\sum *k * \gamma_i * \end{aligned}$$

The $(1 + k)^1$ term arises from the case where only the "1" of each $(1 + \gamma_i)$ is used, as $\sum \epsilon_i = 1$. The $\prod_{i=1}^{p}(1 + \gamma_i)$ arises when only the "1" of each $(1 + k)$ term is used. The correction of "-1" compensates for the case when both these conditions hold. Each other term in the sum contains at least one γ_i and at least one k in either order. Thus

$$\begin{aligned} T(k) &= (1 + k)^1 - W(1 + k) \\ &= (1 + k)^1 - \big(-1 + (1 + k)^1 + (g_1 \cdots g_p) + \sum *k * \gamma_i * \big) \\ &= 1 - (g_1 \cdots g_p) - \sum *k * \gamma_i * \end{aligned}$$

We now prove that T is a contraction. Consider distinct elements $k, \ell \in K$. Letting $\lambda = \mathsf{deg}(k - \ell)$, express these as $k = k' + k''$ and $\ell = \ell' + \ell''$, where $k' = \ell'$ consist of all

terms with degree less than λ. So

$$k' = \sum_{\deg(m)<\lambda} k_m m \quad \text{and} \quad k'' = \sum_{\deg(m)\geq\lambda} k_m m$$
$$\ell' = \sum_{\deg(m)<\lambda} \ell_m m \quad \text{and} \quad \ell'' = \sum_{\deg(m)\geq\lambda} \ell_m m$$

Applying the formula found above for T, and cancelling terms, we have:

$$\begin{aligned} T(k) - T(\ell) &= -\sum *k * \gamma_i * + \sum *\ell * \gamma_i * \\ &= -\sum *(k' + k'') * \gamma_i * + \sum *(\ell' + \ell'') * \gamma_i * \\ &= -\sum *(k'') * \gamma_i * + \sum *(\ell'') * \gamma_i * \end{aligned}$$

where each final sum contains at least one k'' term (or ℓ'' term) and at least one γ_i. Note that $\deg(k - \ell) \geq \lambda$ and the terms not containing k'' or ℓ'' cancel since $k' = \ell'$.

Observe that $\deg(k - \ell) = \deg(k'' - \ell'') = \lambda$. But $\deg(T(k) - T(\ell)) = \deg(\sum *(k'') * \gamma_i * - \sum *(\ell'') * \gamma_i *) \geq \sigma + \lambda$. Thus:

$$\mathsf{d}(T(k), T(\ell)) < 2^{-\sigma} 2^{-\lambda} \;=\; 2^{-\sigma} \cdot \mathsf{d}(k, \ell) \;< \mathsf{d}(k, \ell)$$

Consequently Theorem 3 provides a fixed point for T. □

4 Generalizations

We will apply Newton's Binomial Theorem with $a \neq 0$, and $\deg(x) > 0$:

$$(1 + x)^a \;=\; \sum_{i=0}^{\infty} \binom{a}{i} x^i \tag{1}$$

It is appropriate since 1 and x commute, and is valid since the sequence of partial sums $\sum_{i=0}^{N} \binom{a}{i}(1 + x)^a$ is Cauchy, as $\deg(x) > 0$.

Lemma 6 *Let $k = k' + k''$ where all terms of k' have degree $< d$ and where $\deg(k'') \geq d$. Then*

$$(1 + k' + k'')^{1/r} = 1 + \underline{k}' + \underline{k}''$$

where $\underline{k}'$ depends only on k', and where $\deg(\underline{k}'') \geq \deg(k'')$.

Proof We apply Equation (1) with $a = \frac{1}{r}$ and $x = k' + k''$. Let $\underline{k}'$ denote the terms in this series that involve only powers of k' and no k''. Let $\underline{k}''$ denote the remaining sum. Note that $1 + \underline{k}' = (1 + k')^{1/r}$ so $\underline{k}'$ depends entirely on k'. Furthermore, $\deg(\underline{k}'') \geq \deg(k'')$ since its terms are sums of products of monomials in k'' multiplied by monomials in k'. □

Theorem 7 *Generalization of Theorem 5 where $\#_t(W) = r \neq 0$.*

Proof For $r \neq 0$, Equation (1) gives a map $(1 + k) \mapsto (1 + k)^{1/r}$ where $(1 + k)^{1/r} = 1 + \frac{1}{r}k + \cdots$.

We then have almost the same formula as in the proof of Theorem 5:

$$W\left((1+k)^{\frac{1}{r}}\right) = -1 + (1+k)^1 + \prod_{i=1}^{p}(1+\gamma_i) + \sum *\frac{1}{r}k * \gamma_i *$$

The map $T_r(k) = (1+k) - W\left((1+k)^{\frac{1}{r}}\right)$ again has the property that if $T_r(k) = k$ then $(1+k)^{1/r}$ is a solution to $W(x) = 1$.

Lemma 6 asserts that for our decomposition $k = k' + k''$ we have:

$$(1 + k' + k'')^{1/r} = 1 + \underline{k}' + \underline{k}''$$

where $\underline{k}'$ depends only on k', and where $\mathsf{deg}(\underline{k}'') \geq \mathsf{deg}(k'')$.

The remainder of the proof is essentially the same. □

Remark 8 The following examples illustrate that though we solve the equation $W(t) = 1$ in $R[[M]]^*$, there is no embedding of $G' = (G * \langle t \rangle)/\langle\langle W \rangle\rangle$ in $R[[M]]^*$.

Consider the cyclic group $G = \langle x \rangle$, and the embedding $G \hookrightarrow \mathbb{Z}[[x]]^*$ induced by $x \mapsto (1+x)$. Let $W(t) = xxtt$ (or similarly $xtx^{-1}t$). The group G' is not commutative but $\mathbb{Z}[[x]]$ is commutative. Thus although we solved the equation $W(t) = 1$ in $\mathbb{Z}[[x]]$, there is no embedding of G' in $\mathbb{Z}[[x]]^*$.

Let $W(t) = txt^{-1}x^2tx^3t^{-1}x^4 \cdots tx^n$. Then $\#_t(W) = 1$ but the small-cancellation group $\langle x, t \mid W \rangle$ contains a rank 2 free group for $n \gg 0$.

Let $W(t) = x^{-1}t^3xt^{-2}$. Then G' is the non-residually finite Baumslag-Solitar group. Thus G' cannot embed in any $\mathbb{Z}[[M]]^*$.

Finally, suppose G is the trivial group. Let $W(t) = t^n$ with $n > 0$. Then $G' \cong \mathbb{Z}_n$. However, $\mathbb{Z}[[M]]^*$ is torsion-free, so there is no embedding $G' \hookrightarrow \mathbb{Z}[[M]]^*$. The only solution to $W(t) = 1$ is the trivial solution $t = 1$.

Theorem 9 (System of Equations) *Let $W_1(t_1, \ldots, t_n), \ldots, W_n(t_1, \ldots, t_n)$ be a system of words in the variables $\{t_1, \ldots, t_n\}$ and group elements of $G \subset 1 + K$. Let A be the $n \times n$-matrix with entries $a_{pq} = \#_{t_p}(W_q)$.*

Suppose A is invertible over $\mathbb{Z}$. Then there is a solution to the system $\{W_m = 1\}_{m=1}^n$ in $1 + K$.

Proof By performing Nielsen transformations on the $\{W_m\}$ corresponding to row-reduction operations that transform A to I_n, we obtain a new set of equations where $\#_{t_p}(W_q) = \delta_{pq}$ for $1 \leq p, q \leq n$, and this new system has a solution if and only if the original does. We may thus assume that $A = I_n$.

We now follow the proof of Theorem 5. Let $K^{\oplus n}$ denote the direct product of n copies of K. We metrize $K^{\oplus n}$ using the sum of the metrics on its factors. Consider $T : K^{\oplus n} \to K^{\oplus n}$ where letting $\vec{k} = (k_1, \ldots, k_n)$ we define $T(\vec{k}) = \left(T_1(\vec{k}), \ldots, T_n(\vec{k})\right)$ with $T_m(\vec{k}) = (1 + k_m) - W_m(\vec{1} + \vec{k})$.

Remark 2 continues to hold in this context, so a fixed point $T(\vec{s}) = \vec{s}$ implies $W_m(\vec{1}+\vec{s}) = \vec{1}$ for each $m \in \{1, \ldots, n\}$ as

$$1 = (1 + s_m) - s_m = (1 + s_m) - T_m(\vec{s}) = W_m(\vec{1} + \vec{s})$$

We now verify T is contracting for a suitable metric, and hence has a fixed point.

The formula for $W_m(\vec{1} + \vec{k})$ generalizes by separating the products (for each i) according to whether the 'non-1-term' is used from each $(1 + k_i)$ and the '1-term' is used elsewhere,

but the γ_{qr}-terms are treated together as before. So for $W_m(1+k_1,\dots,1+k_n)$ which is a product of various $(1+k_i)^{\pm 1}$ terms and various $g_{b_i} = (1+\gamma_{b_i})$ constants we have:

$$\begin{aligned} W_m(\vec{1}+\vec{k}) &= (1+\gamma_{b_1})(1+k_{c_1})^{\epsilon_1}\cdots(1+\gamma_{b_\ell})(1+k_{c_\ell})^{\epsilon_\ell} \\ &= \sum_{i=1}^{n}\left((1+k_i)^{\#_{t_i}(W_m)}-1\right) \\ &+ \prod_{i=1}^{\ell}(1+\gamma_{b_i}) \;+\; \sum *k*k* \;+\; \sum *k*\gamma* \\ &= (1+k_m) \;+\; \prod_{i=1}^{\ell}(1+\gamma_{b_i}) - 1 \;+\; \sum *k*k*+\sum *k*\gamma* \end{aligned}$$

The equality at the second line arises by partitioning the terms in the expansion of the product according to the 1's used from the $(1+k_{c_i})$ and $(1+\gamma_{b_i})$ factors. The equality at the third line arises from our hypothesis that $A = I_n$ as it implies that for $p \neq m$, the summand in the second term reduces to $1-1$. Note that the $*k*k*$ terms did not arise in the proof of Theorem 5, but they arise here in the form $*k_i*k_j*$ where $i \neq j$.

Let $\sigma = \min(\mathsf{deg}(\gamma_{ij}))$ and restrict K to consist of those power series having degree $\geq \sigma$.

Let $\vec{k} = \vec{k}'+\vec{k}''$ and $\vec{\ell} = \vec{\ell}'+\vec{\ell}''$ where $\vec{k}' = \vec{\ell}'$ and the first nonzero term of each coordinate of $\vec{k}''$ and $\vec{\ell}''$ are distinct. Consider the m-coordinate of $T(\vec{k}'+\vec{k}'')-T(\vec{\ell}'+\vec{\ell}'')$. Using the expression for W_m above, everything cancels except for terms of the form:

$$\begin{aligned} &-\sum *k''*k''* \;-\; \sum *k''*k'* \;-\; \sum *k''*\gamma* \\ &+\; \sum *\ell''*\ell''* \;+\; \sum *\ell''*\ell'* \;+\; \sum *\ell''*\gamma* \end{aligned}$$

As the degree of any $*$-term above is at least σ, and by definition $\mathsf{deg}(\vec{k}'') = \mathsf{deg}(\vec{\ell}'') = \mathsf{deg}(\vec{k}''-\vec{\ell}'')$, we have:

$$\deg\left(T(\vec{k})-T(\vec{\ell})\right) \;\geq\; \sigma + \min_{i=1}^{n}(\mathsf{deg}(k_i''-\ell_i'')) \;=\; \sigma + \mathsf{deg}(\vec{k}-\vec{\ell}).$$

Consequently $\mathsf{d}(T(\vec{k}),T(\vec{\ell})) \;\leq\; 2^{-\sigma}\mathsf{d}(\vec{k},\vec{\ell})$. □

Theorem 10 *We work in the context of the statement of Theorem 9: If* $\det(A) = r \neq 0$ *then there is a solution to* $\{W_m(t_1,\dots,t_n)=1\}_{m=1}^n$ *in* $1+K$*, where we use the ring* $\mathbb{Z}[\frac{1}{r}][[M]]$.

We give two proofs. The first transforms the system of equations to a new system without group theoretical meaning, but with the same fixed point properties as in the proof of Theorem 9. The second is given after Lemma 15 which provides a trick allowing us to deduce Theorem 10 from Theorem 9.

Proof (First proof of Theorem 10) We can transform the exponent-sum matrix A to I_n by multiplying by a sequence of elementary matrices with coefficients in $\mathbb{Z}[\frac{1}{r}]$. This corresponds to transforming the system of equations $\{W_m(t_1,\dots,t_n)=1\}_{m=1}^n$ to a new system $\{\hat{W}_m(t_1,\dots,t_n)=1\}_{m=1}^n$. For instance, if U_i and U_j are the i-th and j-th equations, then for $d \in \mathbb{Z}[\frac{1}{r}]$, the row-operation of adding d times the i-th equation to the j-th equation corresponds to replacing $U_j = 1$ by $(U_i)^d U_j = 1$. Here letting $U_i = (1+V_i)$ with $V_i \in K$, we apply Equation (1) with $x = V_i$ to obtain $U_i^d = (1+V_i)^d$.

The new set of equations no longer correspond to a set of "relators". However, $\{\hat{W}_m(t_j) = 1\}_{m=1}^n$ has the same solution set as $\{W_m(t_j) = 1\}_{m=1}^n$. Moreover, $\{\hat{W}_m(t_j) = 1\}_{m=1}^n$ satisfies $\#_{t_p}(\hat{W}_q) = \delta_{pq}$ for $1 \le p, q \le n$, where $\#_{t_p}(\hat{W})$ is the coefficient of the t_p monomial of the power series $\hat{W}$. Thus the same argument as for Theorem 9 but using coefficients in $\mathbb{Z}[\frac{1}{r}]$ provides a solution. The crucial hypothesis for the fixed point yielding a solution is that $\#_{t_p}(\hat{W}_q) = \delta_{pq}$, which is what the sequence of "row reductions" achieves. The other supporting points persist (e.g. cancellation of group theoretic constant terms).

We have

$$\hat{W}_m(\vec{1} + \vec{k}) \;=\; (1 + c_m) \;+\; k_m \;+\; \sum *k_i * k_j * \;+\; \sum *k_i * \gamma$$

where c_m is obtained by collecting all terms of degree at least 1 that are independent of $\vec{k}$, where k_m is the nonvanishing term in $\sum_p \#_{t_p}(\hat{W}_m)k_p$, and where the remaining sum consists of multiples of higher degree terms. Here k_i, k_j denote various coordinate elements of $\vec{k}$ but γ denotes arbitrary "coefficients" γ_{b_i} arising from $g_i = (1 + \gamma_{b_i})$.

The transformation $T(\vec{k}) = (\vec{1} + \vec{k}) - \left(\hat{W}_m(\vec{1} + \vec{k})\right)_m$ still has the property that $T(\vec{k}) = \vec{k}$ implies that $\hat{W}_m(\vec{1} + \vec{k}) = 1$ for each m.

The m-th coordinate of $T(\vec{k}) - T(\vec{\ell})$ equals

$$-\sum *k_i * k_j * \;-\; \sum *k_i * \gamma + \sum *\ell_i * \ell_j * \;+\; \sum *\ell_i * \gamma.$$

The contraction property holds using the inequality guided by the algebra comparing $\vec{k} = \vec{k}' + \vec{k}''$ and $\vec{\ell} = \vec{\ell}' + \vec{\ell}''$. As when we distribute, there is cancellation of terms involving only $k_i' = \ell_i'$ so $T(\vec{k}) - T(\vec{\ell})$ equals:

$$\begin{aligned}\sum -\left(*k_i'' * k_j' * + * k_i' * k_j'' * + k_i'' * k_j'' *\right) \;-\; \sum *k_i'' * \gamma \\ + \sum \left(*\ell_i'' * \ell_j' * + * \ell_i' * \ell_j'' * + * \ell_i'' * \ell_j'' *\right) \;+\; \sum *\ell_i'' * \gamma.\end{aligned}$$

Letting K be the ideal consisting of power series with degree $\ge \sigma$, where $\sigma = \min(\mathsf{deg}(\gamma_{ij}))$, we see that:

$$\deg\left(T(\vec{k}) - T(\vec{\ell})\right) \;\ge\; \sigma + \min_{i=1}^{n}(\mathsf{deg}(k_i'' - \ell_i'')) \;=\; \sigma + \mathsf{deg}(\vec{k} - \vec{\ell}).$$

Consequently $\mathsf{d}(T(\vec{k}), T(\vec{\ell})) \le 2^{-\sigma}\,\mathsf{d}(\vec{k}, \vec{\ell})$. □

We now discuss generalizations using characteristic p:

Theorem 11 (Working over $\mathbb{Z}_p$) *Let $W_1(t_1, \ldots, t_n), \ldots, W_n(t_1, \ldots, t_n)$ be a system of words in the variables $\{t_1, \ldots, t_n\}$ and group elements of $G \subset 1 + K \subset \mathbb{Z}_p[[M]]$ for some prime p. Let A be the $n \times n$-matrix whose entries $a_{pq} = \#_{t_p}(W_q)$ are the exponent sums of the variables.*

Suppose A is invertible over $\mathbb{Z}_p$. Then there is a solution to the system $\{W_m = 1\}_{m=1}^n$ in $1 + K$.

Proof This is identical to the proof of Theorem 9 with one small difference. Since elements of $\mathbb{Z}_p$ can be represented by elements of $\mathbb{Z}$, the row operations that transform A to I_n can be represented using $\mathbb{Z}$-matrices. Hence we can diagonalize the coefficient matrix using $\mathbb{Z}$-row operations to obtain a matrix that equals I_n modulo p. Mimicking this in the set of relators yields an equivalent set of equations whose coefficient matrix is I_n modulo p. □

Remark 12 Suppose $G \subset \mathbb{Z}[[M]]$ and the exponent-sum matrix A is invertible over $\mathbb{Q}$. Then A is invertible over $\mathbb{Z}_p$ when $p \nmid \det(A)$. Consider the homomorphism $\mathbb{Z}[[M]] \to \mathbb{Z}_p[[M]]$. Then G is residually in $\mathbb{Z}_p[[M]]$, i.e. each $g \in G$ survives for sufficiently large p. Hence each g survives in $G * \langle t \rangle / \langle\langle W \rangle\rangle$, as a solution to $W = 1$ exists in $\mathbb{Z}_p[[M]]$ when A is invertible modulo p.

5 More abstract treatment of Theorem 10

This section gives an alternate proof of Theorem 10 that is more amenable to generalization and puts more emphasis on the underlying ideas over the computations. In particular, it uses Theorem 9 as a black-box and avoids the need to repeat similar computations.

Definition 13 Let $\mathbb{S} : \mathcal{R} \to \mathcal{R}$ be a ring homomorphism mapping a maximal ideal K into itself. Then $\mathbb{S}(K^2) \subset K^2$, so restricting $\mathbb{S}$ to K we obtain:

$$\mathrm{Jac}(\mathbb{S}) : K/K^2 \longrightarrow K/K^2$$

which we will call the *Jacobian* of $\mathbb{S}$ by analogy. Note that if $\mathcal{R}$ is a commutative power series ring, this agrees with the usual Jacobian.

Definition 14 The *pseudonorm* of $k \in K^{\oplus d}$ is $||k|| := \max_{1 \le i \le d} 2\mathsf{d}(0, k_i)$. The *operator pseudonorm* on $\mathrm{End}(K^{\oplus d})$ is the map $|| \cdot || : \mathrm{End}(K^{\oplus d}) \to \mathbb{R}$ given by $T \mapsto \sup_{k \in K^{\oplus d}} ||T(k)||$, and an endomorphism $T \in \mathrm{End}(K^{\oplus d})$ is a contraction if $||T|| < 1$. This pseudonorm is well-defined since 1 is a uniform upper bound on $||T(k)||$ for each T. The induced topology on $\mathrm{End}(K^{\oplus d})$ is the topology of uniform convergence, which is compatible with composition.

Lemma 15 *Let $\mathcal{R} = R[[t_1, \ldots, t_n]]$. Let $\vec{S} \in \mathcal{R}^{\oplus n}$ be given by $\vec{S} = (S_1, \ldots, S_n)$. Let $\mathbb{S} : \mathcal{R} \to \mathcal{R}$ be induced by $t_i \mapsto S_i(t_1, \ldots, t_n)$. Suppose $\vec{S} = \vec{0} \mod K^{\oplus n}$ where K is the ideal generated by $\{t_1, \ldots, t_n\}$. Suppose the following is invertible over R:*

$$\mathrm{Jac}(\mathbb{S}) : K/K^2 \longrightarrow K/K^2$$

If $T : K^{\oplus d} \to K^{\oplus d}$ is a contraction then $T \circ \mathbb{S}^{\oplus d}$ is also a contraction. Moreover, the set H of all such $\mathbb{S}$ with the composition operation forms a group acting (faithfully) on the set of endomorphisms, and this action stabilizes the subset of contracting endomorphisms.

Proof As $\mathbb{S}(K) \subset K$, the map $\mathbb{S}$ is a well-defined ring homomorphism. Moreover, $\mathrm{Jac}(\mathbb{S})$ induces a map $\mathcal{R} \to \mathcal{R}$ (also abusively denoted by $\mathrm{Jac}(\mathbb{S})$) that agrees to first order with $\mathbb{S}$. Injectivity of $\mathbb{S}$ holds since $\mathbb{S}$ is an isometry:

$$\begin{aligned} \mathsf{d}(0, \mathbb{S}(x)) &= \max\big(\mathsf{d}(0, \mathrm{Jac}(\mathbb{S})(x)), \mathsf{d}(\mathrm{Jac}(\mathbb{S})(x), \mathbb{S}(x))\big) \\ &= \mathsf{d}\big(0, \mathrm{Jac}(\mathbb{S})(x)\big) \;=\; \mathsf{d}(0, x). \end{aligned}$$

From this we deduce the first assertion in the 1-dimensional case $T : K \to K$. Since $\mathrm{Jac}(\mathbb{S})$ is invertible, image$(\mathbb{S})$ surjects onto K/K^2. Since $\left(K/K^2\right)^i$ surjects onto K^i/K^{i+1}, by induction the ideal generated by $\{S_1, \ldots, S_n\}$ surjects onto K/K^i for all i. Surjectivity of $\mathbb{S}$ holds since the source and target of $\mathbb{S}$ are complete with respect to the K-adic topology. Consequently, $\mathbb{S}$ is an automorphism of metrized rings. We conclude that the group H is a subsemigroup of $\mathrm{End}(K)$ and the standard left action of $\mathrm{End}(K)$ on itself restricts to an action of H on contractions. Since $K^{\oplus d} \to K^{\oplus d}$ is a contraction if and only if each of its coordinate functions is contractive, the general case follows. □

Lemma 16 $\mathsf{d} : (T, T') \mapsto ||T - T'||$ *is a complete metric on* $\mathrm{End}(K)$.

Proof Given endomorphisms T, T', T'', the inequality $||T(k)-T''(k)|| \leq ||T(k)-T'(k)||+||T'(k)-T''(k)||$ holds for $k \in K^{\oplus d}$, so by taking suprema we see that d satisfies the triangle inequality. It is nondegenerate because $\sup_{k\in K^{\oplus d}} \mathsf{d}(0, T(k)) = 0$ implies $T(k) = 0$ for all k, which is equivalent to saying T is the 0 endomorphism.

For a Cauchy sequence $\{T_\alpha\}_\alpha \subset \mathrm{End}(K)$, define $T(k) := \lim_\alpha(T_\alpha(k))$. From the definitions, T is an endomorphism and $(\lim_\alpha T_\alpha)(k) = T$. □

Remark 17 $T \in \mathrm{End}(K)$ is a contraction if and only if $||T|| < 1$. Thus if $T, T' \in \mathrm{End}(K)$ satisfy $||T|| < 1$ and $||T' - T|| < 1$, then $||T'|| < 1$ so T' is a contraction. Moreover, the pseudonorm is invariant under extension of scalars, hence enlarging the coefficient ring preserves contractions.

Below W denotes a system of equations $\{W_m(t_1, \ldots, t_n) = 1\}_{m=1}^n$ arising from words in $G * \langle t_1, \ldots, t_n\rangle$. We use $\widehat{W}$ to denote general equations in the power series ring. We often precompose with $\mathbb{S} \in \mathrm{End}(K)^{\oplus n}$ to obtain a new system $\widehat{W} \circ \mathbb{S}$ which denotes $\{\widehat{W}_m(\mathbb{S}_m(t_1), \ldots, \mathbb{S}_m(t_n)) = 1\}_{m=1}^n$.

Lemma 18 *Given a system of equations* $\widehat{W}$ *in* $K^{\oplus n} \subset \mathcal{R}^{\oplus n}$, *the mapping* $\{t_i \mapsto \widehat{W}_m(t_1, \ldots, t_n)\}_{m=1}^n$ *induces a morphism* $\mathcal{R} \to \mathcal{R}$, *whose Jacobian will be denoted by* $\mathrm{Jac}(\widehat{W})$. *If the endomorphism*

$$T(\vec{k}; \widehat{W}) := \left(\vec{1} + \mathrm{Jac}(\widehat{W})(k)\right) - \widehat{W}(\vec{1} + \vec{k})$$

of $\mathcal{R}^{\oplus n}$ *is a contraction and* $\mathbb{S} \in \mathrm{End}(K)^{\oplus n}$ *is an isometry, then* $T(\vec{k}; \widehat{W} \circ \mathbb{S})$ *is a contraction.*

Proof Composing $T(\text{-}; \widehat{W})$ with $\mathbb{S}$ we obtain $T(\mathbb{S}(\text{-}); \widehat{W})$ which is a contraction by Lemma 15. But

$$||T(\mathbb{S}(\text{-}); \widehat{W}) - T(\text{-}; \widehat{W} \circ \mathbb{S})|| = ||\mathrm{Jac}(\widehat{W} \circ \mathbb{S})(\text{-}) - \mathrm{Jac}(\widehat{W})(\mathbb{S}(\text{-}))|| \leq \frac{1}{2} < 1.$$

Applying Remark 17 coordinate by coordinate completes the proof. □

Proof (Second proof of Theorem 10) Combining the computation of the proof of Theorem 9 with Lemmas 15 and 18, we see that for $\mathbb{S} \in \mathrm{End}(K)^{\oplus n}$, the map $T(\text{-}; \widehat{W} \circ \mathbb{S}) := \left(\vec{1} + \mathrm{Jac}(\widehat{W} \circ \mathbb{S})(\text{-})\right) - \widehat{W} \circ \mathbb{S}(\text{-})$ is also a contraction. Hence in the special case where W is a set of words in $G * \langle t_1, ..., t_n\rangle$ whose matrix $A := \left(\#_{t_p}(W_q)\right)$ has $\det(A) = 1$, the system of equations $\widehat{W} \circ \mathbb{S}(\vec{1} + \vec{k}) = \vec{1}$ has a unique solution. This implies that if $\widehat{W}'$ is a system of equation for which there exist $\mathbb{S} \in \mathrm{Aut}(K)$ such that the system of equations $\widehat{W}' \circ \mathbb{S}$ comes from a system of words W'', then $\widehat{W}'(\vec{1} + \vec{k}) = \vec{1}$ has a unique solution: if $\vec{k}$ is a fixed point of the contraction induced by $\widehat{W''}$, the solution is $\mathbb{S}^{-1}(\vec{k})$.

In the general case where $\det(A) = r$, we will proceed by finding a transformation $\mathbb{S}$ such that precomposing with $\mathbb{S}$ reduces the problem to the case $\det(A) = 1$. To produce such an $\mathbb{S}$, we first enlarge the coefficient ring to $R[\sqrt[n]{1/r}\,] \otimes \mathbb{Q}$.

Equation (1) gives a power series for $(1 + t_i)^{\sqrt[n]{1/r}}$, and the mapping $\left\{t_i \mapsto (1 + t_i)^{\sqrt[n]{1/r}} - 1\right\}_{i=1}^n$ induces a morphism $\mathbb{S} \in \mathrm{End}(K)^{\oplus n}$. We precompose with $\mathbb{S}$ to get a new system of equation $\widehat{W}' := \widehat{W} \circ \mathbb{S}$ and observe that the matrix $A' := \left(\#_{t_p}(\widehat{W}'_q)\right)$ has $\det(A') = \det(A) \cdot \frac{1}{r} = 1$. We now know that $T(\text{-}, \widehat{W}')$ is a contraction, hence that $T(\text{-}, \widehat{W})$ is a contraction when the coefficient ring is $R[\sqrt[n]{1/r}\,] \otimes \mathbb{Q}$, and thus by Remark 17 the original map T is a contraction. □

Remark 19 We chose $\mathbb{S}$ because $\det(\mathrm{Jac}(\mathbb{S})) = \frac{1}{r}$. It plays the role of a multivariate generalization of the transformation used in Theorem 7.

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